MT DATA Feed API

Version 1.1

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MT Data Feed API Documentation

Overview

The MT Data Feed API allows you to receive real-time market data from various exchanges and segments using WebSocket technology. WebSocket maintains a persistent connection, enabling efficient live data streaming. All communication (requests and responses) is in JSON format. To use the live data feed, implement a WebSocket client in your preferred programming language.

WebSocket Endpoints

Live: wss://feedapi.com

UAT (Testing): **ws://115.242.15.134:19101**

General Packet Structure

Each message to and from the API uses the following format:

```
{
  "Type": "string",
  "Data": "object"
}
```

API Requests

Login Request

```
{
  "LoginId": "string",
  "Password": "string"
}
```

Type: Login		Data: Login Request Object
LoginId	String	Login Id received for API Provider
Password	String	Password received for API Provider

Login Response

```
{
  "Error": "string",
  "LoginId": "string",
  "Version": "string",
  "Xchgs": "string"
}
```

Logout Request

```
{
  “string”
 }
```

Type: Logout	Data: Any String Message

Token Request

```
{
   "SubType": "boolean",   // true = Subscribe, false = Unsubscribe
   "FeedType": "integer",  // 1 = MarketData, 2 = Depth
   "quotes": [
     {
       "Xchg": "string",    // Exchange name
       "Tkn": "string"      // Token ID
       "Symbol": "string"      // Script Name
     }
   ]
}
```

Type: TokenRequest		Data: Token Request Object
SubType	Boolean	True=Subscribe, false=Unsubscribe
FeedType	Int	1=MarketData, 2=Depth, 3=SnapQuote,4=Greeks
Quotes	Array	Ref TALBE QUOTE

TABLE: QUOTE		
Xch	String	Exchange Name
Tkn	String	TokenNo as Script Identifier
Symbol	String	Script Name

API Responses

Login Response

Type: Login		Data: Login Response Object
Error	String	Error message on Failure, blank on Success
LoginId	String	LoginId as requested on Login Request
Version	String	Current API Version
Xchgs	String	List of allowed Exchanges as comma separated

Logout Response

Type: Logout		Data: Logout Message if any
Data	String	In API Packet Data will be received with string rather than Object

Feed Status Response

Type: FeedStatus		Data: FeedStatus Array
Xchg	String	Exchange Name
Status	Int	1=Connected, 0=Disconnected

Information Response

Type: Info		Data: Information Object
InfoType	String	Error, Warning, XchgMsg, BrkMsg
InfoMsg	String	Message to be displayed

To keep the connection live one must send Information Packet to the Server at every minute by setting InfoType="HB"(heartbeat) so server identify the connection as live. One needs to send this only when no activity is done from Client End.

Market Data Response

Type: MarketData		Data: MarketData Object
Xchg	String	Exchange Name
Tkn	String	TokenNo Script Identity
LTP	Decimal	Last Traded Price
LTQ	Decimal	Last Traded Quantity
LUT	String	Last Update Time in format “HH:mm:ss”
LTT	String	Last Trade Time in format “HH:mm:ss”
ATP	Decimal	Average Trade Price
BQ	Decimal	Bid Quantity (Top Bid Qty)
BP	Decimal	Bid Price (Top Bid Price)
SQ	Decimal	Ask Quantity (Top Ask Qty)
SP	Decimal	Ask Price (Top Ask Price)
TBQ	Decimal	Total Bid Quantity
TSQ	Decimal	Total Ask Quantity
TTQ	Decimal	Total Traded Quantity (Volume)
TTV	Decimal	Total Traded Value
OI	Decimal	Current Open Interest
O	Decimal	Open Price
H	Decimal	High Price
L	Decimal	Low Price
C	Decimal	Close Price
DPRL	Decimal	Lower Price Range of Day
DPRH	Decimal	Higher Price Range of Day

Depth Response

Type: Depth		Data: Depth Object
Xchg	String	Exchange Name
Tkn	String	Token No, Script Identity
Depths	Array	Depth Object Refer Table DEPTH

TABLE: DEPTH		
BO	Decimal	No of Bids
BP	Decimal	Bid Price
BQ	Decimal	Bid Qty
SQ	Decimal	Ask Qty

SP	Decimal	Ask Price
SO	Decimal	No of Asks

Index Data Response

Type: IndexData	Data: IndexData Object Array
	Refer Table INDEXDATA

TABLE: INDEXDATA		
Symbol	String	Index Name
Price	Decimal	Index Value
O	Decimal	Todays Open Value of Index
H	Decimal	Todays High Value of Index
L	Decimal	Todays Low Value of Index
C	Decimal	Previous Days Close Value of Index

Greeks Mkt Data Response

Type: Greek		Data: GreekData Object
Xchg	String	Exchange Name
Tkn	String	TokenNo Script Identity
Symbol	String	Full Symbol Description
LTP	Decimal	Last Traded Price
LTQ	Decimal	Last Traded Quantity
LUT	String	Last Update Time in format “HH:mm:ss”
LTT	String	Last Trade Time in format “HH:mm:ss”
ATP	Decimal	Average Trade Price
BQ	Decimal	Bid Quantity (Top Bid Qty)
BP	Decimal	Bid Price (Top Bid Price)
SQ	Decimal	Ask Quantity (Top Ask Qty)
SP	Decimal	Ask Price (Top Ask Price)
TBQ	Decimal	Total Bid Quantity
TSQ	Decimal	Total Ask Quantity
TTQ	Decimal	Total Traded Quantity (Volume)
TTV	Decimal	Total Traded Value
OI	Decimal	Current Open Interest
O	Decimal	Open Price
H	Decimal	High Price

L	Decimal	Low Price
C	Decimal	Close Price
DPRL	Decimal	Lower Price Range of Day
DPRH	Decimal	Higher Price Range of Day
SpotPrice	Decimal	Spot Price for Underlying Symbol
IV	Decimal	Volatility
Delta	Decimal	Current Delta for this Script
Gamma	Decimal	Current Gamma for this Script
Theta	Decimal	Current Theta for this Script
Vega	Decimal	Current Vega for this Script
Rho	Decimal	Current Rho for this Script
Depths	Arrya	Depth Object Refer Table DEPTH

Opt Chain Response

Type: OChain		Data: OptChain Object
SubType	Boolean	
Xchg	String	Exchange Name
Symbol	String	Symbol Name like NIFTY, BANKNIFTY, SENSEX
Expiry	String	Expiry Date of this Symbol
SpotPrice	Decimal	Live underlying price
VIX	Decimal	India Volatility Index
MaxPain	Decimal	Max Pain point
CallVolTot	Decimal	Total call option volume
PutVolTot	Decimal	Total put option volume
CEIVTotal	Decimal	Total IV of all Call options
PEIVTotal	Decimal	Total IV of all Put options
IVSkew	Decimal	PEIVTotal - CEIVTotal
CEOITot	Decimal	Total open interest for Calls
PEOITot	Decimal	Total open interest for Puts
PCR	Decimal	Put/Call OI ratio
OptChain	Arrya	Option Chain TABLE

Below Table contains All Call and Put Data for this Strike Price.

TABLE: Option Chain		
Strike	Decimal	Call Put Data for this Strike Price
OIPCR	Decimal	Individual PCR for this Strike
IVSkew	Decimal	IV Difference (Put IV – Call IV)
CETkn	String	Token or ID for the CE contract
CLTP	Decimal	Last traded price of Call
CBidQ	Decimal	Call Bid Quantity
CBidP	Decimal	Call Bid Price
CAskP	Decimal	Call Ask Price
CAskQ	Decimal	Call Ask Quantity
COI	Decimal	Call Open Interest
CVol	Decimal	Call Traded Volume
CBuyer	Decimal	Call Buyer Quantity
CSeller	Decimal	Call Seller Quantity
CHVNBids	Decimal	Highest bid Volume cluster node size
CHVNBidPrice	Decmial	Price at which bid node is highest
CHVNAsks	Decimal	Highest ask Volume cluster node size
CHVNAskPrice	Decmial	Price at which ask node is highest
CIV	Decimal	Call Implied Volatility
CTheta	Decimal	Call Theta
CDelta	Decimal	Call Delta
CGamma	Decimal	Call Gamma
CVega	Decimal	Call Vega
CEVWAP	Decimal	Call VWAP
CEAncVWAP	Decimal	Anchor VWAP from low or open
CETrendScore	Decimal	Trend score for CE
CEPremDivergance	Decimal	Premium divergence score for CE
PETkn	String	Put TokenNo
PLTP	Decimal	Last traded price of Put
PBidQ	Decimal	Put Bid Quantity
PBid	Decimal	Put Bid Price
PAsk	Decimal	Put Ask Price
PAskQ	Decimal	Put Ask Quantity
POI	Decimal	Put Open Interest
PVol	Decimal	Put Traded Volume
PBuyer	Decimal	Put Buyer Quantity
PSeller	Decimal	Put Seller Quantity
PHVNBids	Decimal	Highest bid Volume cluster node size

PHVNBidPrice	Decimal	Price at which bid node is highest
PHVNAsks	Decimal	Highest ask Volume cluster node size
PHVNAskPrice	Decimal	Price at which ask node is highest
PIV	Decimal	Put Implied Volatility
PTheta	Decimal	Put Theta
PDelta	Decimal	Put Delta
PGamma	Decimal	Put Gamma
PVega	Decimal	Put Vega
PEVWAP	Decimal	Put VWAP
PEAncVWAP	Decimal	Anchor VWAP from low or open
PETrendScore	Decimal	Trend score for PE
PEPremDivergance	Decimal	Premium divergence score for PE

CandleBar Response

Type: CandleBar		Data: OHLCBar Object
Xchg	String	Exchange Name
Symbol	String	Symbol Name like NIFTY, BANKNIFTY, SENSEX
Expiry	String	Expiry Date of this Symbol
VIX	Decimal	India Volatility Index
ohlcBars	Arrya	OHLC Bar TABLE

TABLE: OHLC BAR		
O	Decimal	Open Price for this Bar
H	Decimal	High Price for this Bar
L	Decimal	Low Price for this Bar
C	String	Close Price for this Bar
Vol	Decimal	No of Volume for this time interval
Time	String	Time in “HH:mm” format
OI	Decimal	Current OI at this Bar
OIChng	Decimal	OI Change form prev Bar
VWAP	Decimal	VWAP at this BAR
AncVWAP	Decimal	Anchor VWAP at this BAR
CVD	Decimal	Seller-Buyer Difference at this Bar

Message Types

Type	Description
Login	Authenticate and start the session
Logout	Logout and close the session
TokenRequest	Subscribe/Unsubscribe to instruments
Error	Error message for invalid requests
FeedStatus	Get the connection status of an exchange
MarketData	Receive real-time price and volume information
Greek	Receive real-time price market data will Greeks
Depth	Receive market depth (Bid/Ask data)
IndexData	Get major index values like NIFTY50, SENSEX
Info	General informational messages
OChain	OptionChain this message type is used for Request & Response both.
CandleBar	OHLC Candle Bar Records

Examples

Login Request

```
{
  "Type": "Login",
  "Data": {
    "LoginId": "TEST",
    "Password": "testME"
  }
}
```

Logout Request

```
{
  "Type": "Logout",
  "Data": "logout"
}
```

Token Request

```
{
  "Type": "TokenRequest",
  "Data": {
    "SubType": true,
    "FeedType": 1,
    "quotes":[
      {
        "Xchg":"NSECM",
        "Tkn":"2885",
        "Symbol":"RELIANCE"
      },
      {
        "Xchg":"NSEFO",
        "Tkn":"57133",
        "Symbol":"NIFTY"
      }
    ]
  }
}
```

OptionChain Request

```
{
   "Type": "OChain",
   "Data": {
      "SubType": true,
      "Xchg":"NSEFO",
      "Symbol":"NIFTY",
      "Expiry":"03-07-2025"//Option’s Expiry Date
   }
}
```